

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2024/2025

JANUARY EXAMINATION

AMIS1012 ETHICS IN COMPUTING

TUESDAY, 7 JANUARY 2025

TIME: 9.00 AM – 11.00 AM (2 HOURS)

DIPLOMA IN COMPUTER SCIENCE

DIPLOMA IN INFORMATION TECHNOLOGY

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

AMIS1012 ETHICS IN COMPUTING**Question 1**

- a) Distinguish between morals and ethics in the context of IT, and provide an example of each to support your discussion. (8 marks)
- b) Define cybercrime. (2 marks)
- c) Describe **TWO (2)** factors that contribute to the rise of computer-related vulnerabilities. (6 marks)
- d) Explain **THREE (3)** examples of active security attacks. (9 marks)

[Total: 25 marks]

Question 2

Cybertron Sdn. Bhd. is a cloud storage provider. Recently, the risk assessment team identified serious security vulnerabilities in their systems and networks. Attackers exploited weak access controls and unmonitored employee practices. In response, Cybertron Sdn. Bhd. has decided to enhance its cybersecurity practices by implementing robust IT security policies.

- a) Identify the purpose of an IT Security Policy. (1 mark)
- b) Provide **FOUR (4)** examples of IT Security Policy that Cybertron Sdn. Bhd. can implement. Explain the importance of each example of IT Security Policy. (12 marks)
- c) Based on the Confidentiality, Integrity, and Availability (CIA) Triad, assess the **Confidentiality** principle and the **Availability** principle that can be adopted at Cybertron Sdn. Bhd. Support your explanation with an example for each principle. (8 marks)
- d) Discuss how the **FOUR (4)** pillars of Trustworthy Computing can be applied at Cybertron Sdn. Bhd. to enhance the organisation's IT security. (4 marks)

[Total: 25 marks]

Question 3

- a) Determine **TWO (2)** goals to be accomplished using software testing to ensure high-quality software. (4 marks)

AMIS1012 ETHICS IN COMPUTING**Question 3 (Continued)**

- b) Examine the differences between black-box and white-box testing methods in software development, and provide a scenario for each testing method where it should be applied. (8 marks)
- c) Explain the purpose of the Plan-Do-Check-Act (PDCA) cycle in quality assurance and describe its stages. (10 marks)
- d) Identify the impact of discovering a flaw later in the production process. (3 marks)

[Total: 25 marks]

Question 4

Cadiz Sdn. Bhd. has recently launched a new AI-powered online shopping portal. Soon after the launch, they discovered that a competitor was using their portal's unique design features and branding elements without permission. Upon investigation, Cadiz Sdn. Bhd. found that their intellectual property (IP) protections were insufficient.

- a) Determine the importance of protecting intellectual property rights (IPR) for Cadiz Sdn. Bhd. Discuss how IPR contributes to competitive advantage. (5 marks)
- b) Interpret the following types of intellectual property protections relevant to Cadiz Sdn. Bhd. and discuss how each could be applied to protect their newly launched AI-powered online shopping portal.
- (i) Patents (3 marks)
 - (ii) Copyrights (3 marks)
 - (iii) Trademarks (2 marks)
 - (iv) Trade Secrets (2 marks)
- c) Cyber squatting is one of the issues related to intellectual property. Explain to Cadiz Sdn. Bhd. the following types of cybersquatting that they need to be aware of.
- (i) Reverse-cybersquatting (3 marks)
 - (ii) Typosquatting (3 marks)
 - (iii) Name jacking (2 marks)

AMIS1012 ETHICS IN COMPUTING**Question 4 c) (Continued)**

(iv) Identity theft

(2 marks)

[Total: 25 marks]